MSG No. 64.479 C_amca [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0, \frac{3}{4}, 0\right]$	`[1,6,26,29]`
2	$\left[0, \frac{1}{4}, 0\right]$	`[2,5,25,30]`
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,8,28,31]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,7,27,32]`
5	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[9,14,18,21]`
6	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[10,13,17,22]`
7	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[11,16,20,23]`
8	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[12,15,19,24]`

Table 2: Wyckoff site: `8b`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[1,13,22,26]`
2	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2,14,21,25]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,15,24,28]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,16,23,27]`
5	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[5,9,18,30]`
6	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[6,10,17,29]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[7,11,20,32]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[8,12,19,31]`

Table 3: Wyckoff site: `8c`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,15,16]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,21,22]`
6	$\left[0, 0, \frac{1}{2}\right]$	`[19,20,23,24]`
7	$\left[0, \frac{1}{2}, 0\right]$	`[25,26,29,30]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[27,28,31,32]`

Table 4: Wyckoff site: `8d`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,21,22]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[3,4,23,24]`
3	$\left[\frac{3}{4}, 0, 0\right]$	`[5,6,17,18]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[7,8,19,20]`
5	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[9,10,29,30]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,31,32]`
7	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[13,14,25,26]`
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,16,27,28]`

Table 5: Wyckoff site: `8e`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[0, y, \frac{1}{4}\right]$	`[1,6,19,24]`
2	$\left[0, -y, \frac{3}{4}\right]$	`[2,5,20,23]`
3	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[3,8,17,22]`
4	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[4,7,18,21]`
5	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[9,14,27,32]`
6	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[10,13,28,31]`
7	$\left[0, y+\frac{1}{2}, \frac{1}{4}\right]$	`[11,16,25,30]`
8	$\left[0, \frac{1}{2}-y, \frac{3}{4}\right]$	`[12,15,26,29]`

Table 6: Wyckoff site: `8f`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	`[1,3,22,24]`
2	$\left[\frac{1}{4}, -y, \frac{3}{4}\right]$	`[2,4,21,23]`
3	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[5,7,18,20]`
4	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	`[6,8,17,19]`
5	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[9,11,30,32]`
6	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[10,12,29,31]`
7	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[13,15,26,28]`
8	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[14,16,25,27]`

Table 7: Wyckoff site: `16g`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x, \frac{3}{4}, 0\right]$	`[1,26]`
2	$\left[x, \frac{1}{4}, 0\right]$	`[2,25]`
3	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{2}\right]$	`[3,28]`
4	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{2}\right]$	`[4,27]`
5	$\left[-x, \frac{1}{4}, 0\right]$	`[5,30]`
6	$\left[-x, \frac{3}{4}, 0\right]$	`[6,29]`
7	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[7,32]`
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[8,31]`
9	$\left[x+\frac{1}{2}, \frac{1}{4}, 0\right]$	`[9,18]`
10	$\left[x+\frac{1}{2}, \frac{3}{4}, 0\right]$	`[10,17]`
11	$\left[-x, \frac{1}{4}, \frac{1}{2}\right]$	`[11,20]`
12	$\left[-x, \frac{3}{4}, \frac{1}{2}\right]$	`[12,19]`

continued ...

Table 7

No.	position	mapping
13	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ 0\right]$	[13,22]
14	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[14,21]
15	$\left[x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[15,24]
16	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[16,23]

Table 8: Wyckoff site: 16h, site symmetry: 2..

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2]
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[3,4]
3	$[-x,\ 0,\ 0]$	[5,6]
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[7,8]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[9,10]
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[11,12]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[13,14]
8	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[15,16]
9	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[17,18]
10	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[19,20]
11	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[21,22]
12	$\left[x,\ 0,\ \frac{1}{2}\right]$	[23,24]
13	$\left[x,\ \frac{1}{2},\ 0\right]$	[25,26]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[27,28]
15	$\left[-x,\ \frac{1}{2},\ 0\right]$	[29,30]
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[31,32]

Table 9: Wyckoff site: 16i, site symmetry: m..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,6]
2	$[0,\ -y,\ -z]$	[2,5]
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[3,8]
4	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[4,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9,14]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10,13]
7	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,16]
8	$\left[0,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12,15]
9	$\left[\frac{1}{2},\ y,\ z\right]$	[17,22]
10	$\left[\frac{1}{2},\ -y,\ -z\right]$	[18,21]
11	$\left[0,\ y,\ \frac{1}{2}-z\right]$	[19,24]
12	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	[20,23]
13	$\left[0,\ y+\frac{1}{2},\ z\right]$	[25,30]
14	$\left[0,\ \frac{1}{2}-y,\ -z\right]$	[26,29]
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27,32]
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[28,31]

Table 10: Wyckoff site: `16j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,22]`
2	$\left[\frac{1}{4},\ -y,\ -z\right]$	`[2,21]`
3	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-z\right]$	`[3,24]`
4	$\left[\frac{1}{4},\ -y,\ z+\frac{1}{2}\right]$	`[4,23]`
5	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[5,18]`
6	$\left[\frac{3}{4},\ y,\ z\right]$	`[6,17]`
7	$\left[\frac{3}{4},\ -y,\ z+\frac{1}{2}\right]$	`[7,20]`
8	$\left[\frac{3}{4},\ y,\ \frac{1}{2}-z\right]$	`[8,19]`
9	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z\right]$	`[9,30]`
10	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -z\right]$	`[10,29]`
11	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,32]`
12	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,31]`
13	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ -z\right]$	`[13,26]`
14	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	`[14,25]`
15	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[15,28]`
16	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[16,27]`

Table 11: Wyckoff site: `16k`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,24]`
2	$\left[x,\ -y,\ \frac{3}{4}\right]$	`[2,23]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{4}\right]$	`[3,22]`
4	$\left[\frac{1}{2}-x,\ -y,\ \frac{3}{4}\right]$	`[4,21]`
5	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[5,20]`
6	$\left[-x,\ y,\ \frac{1}{4}\right]$	`[6,19]`
7	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[7,18]`
8	$\left[x+\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[8,17]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,32]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,31]`
11	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,30]`
12	$\left[-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[12,29]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[13,28]`
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[14,27]`
15	$\left[x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[15,26]`
16	$\left[x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[16,25]`

Table 12: Wyckoff site: `32l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`

continued ...

Table 12

No.	position	mapping
5	$[-x, -y, -z]$	[5]
6	$[-x, y, z]$	[6]
7	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2}, y, \frac{1}{2}-z\right]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$	[9]
10	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[10]
11	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[11]
12	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-y, -z\right]$	[13]
14	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z\right]$	[14]
15	$\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[15]
16	$\left[x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[16]
17	$\left[x+\frac{1}{2}, y, z\right]$	[17]
18	$\left[x+\frac{1}{2}, -y, -z\right]$	[18]
19	$\left[-x, y, \frac{1}{2}-z\right]$	[19]
20	$\left[-x, -y, z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x, -y, -z\right]$	[21]
22	$\left[\frac{1}{2}-x, y, z\right]$	[22]
23	$\left[x, -y, z+\frac{1}{2}\right]$	[23]
24	$\left[x, y, \frac{1}{2}-z\right]$	[24]
25	$\left[x, y+\frac{1}{2}, z\right]$	[25]
26	$\left[x, \frac{1}{2}-y, -z\right]$	[26]
27	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[28]
29	$\left[-x, \frac{1}{2}-y, -z\right]$	[29]
30	$\left[-x, y+\frac{1}{2}, z\right]$	[30]
31	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[31]
32	$\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[32]